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Unit 2 Lab A

Do the programming exercise listed below. Use the following naming convention for the filename that you turn in: yourlastname_Unit2laba.cpp.

Use Attached Current Code Snippet refueling.cpp as a template to begin this exercise.

Design. You will create using an IPO diagram to design and correct this program. See IPO diagram attached to this assignment. Under INPUTS, list everything the program will need to produce the result. Under OUTPUTS, list what the program is going to produce. Under the PROCESS heading, list the detailed pseudocode steps that need to occur to produce the program's output. This includes formulas your program will need.

Coding. Use your pseudocode to write your programs.

Requirements:

- ☒ (1) Collect from the user estimated miles you will travel and how much fuel your car will hold.
- ☒ (2) Input must allow for decimals to be entered
- ☒ (3) Create two displays for distance range for miles - one with a rounded to the nearest full mile and the other showing all decimal places.
- ☒ (4) Fix all the errors in the existing sample code but do not just get new code from AI or the internet
- ☒ (5) Put a header comment in the code with your programmer name, description of the program and the date the program was last changed.
- ☒ (6) Your program should conform to the Programming Style Requirements as listed in Blackboard under Course Content Turn in your source (.cpp) file and IPO to Blackboard (attach all files before pressing SUBMIT).

*have notes from 8/26

collect

① Estimated mi

② Fuel capacity

convert

① Point-float input

output

① nearest whole mile

② full sig-fig mile

IPO Diagram

Input	Process	Output
• Estimated miles - (float) • Fuel capacity - (float)	• Calculate & Store Range - $\text{range} = \frac{\text{miles}}{\text{capacity}}$ • (float) capacity	• Display 1 - rounded to nearest mile • Display 2 - full decimal values

Input	Process	Output